
Subject Review

Obsessive Compulsive Disorder: A Clinical Update

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Recent epidemiologic studies have indicated that obsessive compulsive disorder is a fairly common psychiatric condition. In this article, obsessive compulsive disorder, its subtypes, and epidemiologic features are described. The common obsessions and compulsions are discussed, as are the comorbid conditions. Currently, etiologic hypotheses revolve around serotonergic and dopaminergic neurotransmitter systems. Management of patients with obsessive compulsive disorder involves education, pharmacotherapy, and behavior therapy, and recent advances in such treatments offer patients hope. Psychotherapy has not proved useful. In refractory cases, psychosurgical intervention remains an option.

Obsessive compulsive disorder (OCD) is a common, potentially disabling anxiety disorder. Recently, public attention has been drawn to this condition by media presentations and informative books.¹⁻³ These communiqués have informed the public about the nature of OCD and the treatment of patients with this condition; thus, the self-referral rate of patients has increased.

OCD has become more manageable because of recent advances in behavior therapy and psychopharmacotherapy. Herein, updated information on the description, epidemiologic features, and causes of OCD is presented. Comorbidity is discussed, as are the available treatment options. The combination of education, pharmacotherapy, and behavior therapy is emphasized. Psychosurgical intervention is described as a last-resort option.

DESCRIPTION

OCD has been described by many authors throughout the centuries. Freud's description of Rat Man⁴ addresses the psychodynamics and defense mechanisms of a patient with OCD. Recently, the *Diagnostic and Statistical Manuals of Mental Disorders*, published by the American Psychiatric Association, have included either OCD or obsessive compulsive neurosis in every edition. The revised third edition (DSM-III-R) categorizes OCD as a type of anxiety disorder.⁵ Thus, OCD is grouped with other conditions that have anxiety

or phobia as a core symptom, including panic disorder with and without agoraphobia, agoraphobia without history of panic disorder, social phobia, simple phobia, posttraumatic stress disorder, generalized anxiety disorder, and anxiety disorder not otherwise specified.

The criteria for OCD from the DSM-III-R are listed in Table 1. Patients with OCD suffer from obsessions—unwelcome ideas, thoughts, urges, or images that repeatedly intrude on their consciousness and seem to the patients, at least initially, to be senseless and unnecessary. Such patients recognize that their obsessions are a product of their own minds and not impositions from outside, such as is commonly observed in patients with psychosis. Compulsions, cognitive as well as physical, are activities performed ritualistically, usually in an attempt to decrease the anxiety caused by an obsession. Most adult patients realize that their obsessional thoughts and compulsive behavior are excessive or unreasonable. Some patients, however, “lose sight” of the excessiveness of their thoughts and behavior when their obsessions evolve into “overvalued ideas.” An overvalued idea may be a delusion in that patients believe that their obsession is reasonable and vital to their well-being or the well-being of others. A final element in the diagnosis of OCD based on the DSM-III-R is that the obsessions or compulsions cause considerable distress, are time-consuming, and therefore substantially interfere with the person's life. The DSM-III-R does not fully address the feared disasters, also known as catastrophic thinking, which are a component for some patients. In such patients, anxiety is gener-

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Table 1.—**Diagnostic Criteria for Obsessive Compulsive Disorder, Based on *Diagnostic and Statistical Manual of Mental Disorders (Revised Third Edition)***

- A. Either obsessions or compulsions
- Obsessions:*
1. Recurrent and persistent ideas, thoughts, impulses, or images that are experienced, at least initially, as intrusive and senseless—for example, a parent's repeated impulses to kill a loved child or a religious person's recurrent blasphemous thoughts
 2. Person attempts to ignore or suppress such thoughts or impulses or to neutralize them with some other thought or action
 3. Person recognizes that the obsessions are the product of his or her own mind, not imposed from without (as in thought insertion)
 4. If another axis I disorder is present, the content of the obsession is unrelated to it—for example, the ideas, thoughts, impulses, or images are not about food in the presence of an eating disorder, about drugs in the presence of a psychoactive substance use disorder, or guilty thoughts in the presence of a major depression
- Compulsions:*
1. Repetitive, purposeful, and intentional behaviors that are performed in response to an obsession, according to certain rules, or in a stereotyped fashion
 2. Behavior is designed to neutralize or to prevent discomfort or some dreaded event or situation; however, either the activity is not connected in a realistic way with what it is designed to neutralize or prevent or it is clearly excessive
 3. Person recognizes that his or her behavior is excessive or unreasonable (this may not be true for young children; it may no longer be true for people whose obsessions have evolved into overvalued ideas)
- B. Obsessions or compulsions cause marked distress, are time-consuming (take more than an hour a day), or significantly interfere with the person's normal routine, occupational functioning, usual social activities, or relationships with others

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ated by the anticipation of misfortune that they fear will occur if they confront a stimulus that results in an obsession and the compulsion is not performed. Sometimes, catastrophic thinking is identical to obsessional thinking, but often it differs, and clinicians must ask patients what they fear will occur, in detail, if a compulsion is not performed. For example, patients who wash their hands excessively because of an obsession with cleanliness may not say that their greatest fear is that they will transfer germs to their children, an outcome that would cause their children to become severely ill or die. Detailed knowledge of a patient's obsessions, compulsions, and catastrophic thinking is necessary for proper behavioral management.

Numerous obsessions and compulsions associated with OCD are shown in Tables 2 and 3.⁶ Frequently, patients with

OCD are categorized on the basis of the rituals they perform. Stern and Cobb⁷ surveyed 45 patients with OCD to determine the frequency of occurrence of obsessive compulsive symptoms. (Some patients had more than one compulsion; therefore, percentages total more than 100%.) The following were the most common rituals. (1) Cleaning (51% of patients), caused by a fear of contamination by dirt, germs, urine, feces, chemicals, or any other repugnant substance. (2) Repeating (40%), which occurs when patients repeat a ritual a certain number of times to prevent a catastrophe. Frequently, such patients have "magic numbers" that dictate the number of times the compulsion must be repeated to ensure safety. (3) Completing compulsions (11%), a type of repeating ritual in which patients believe that a behavior must be completed in an exact and often complicated fashion in order for the task to prevent some unrealistically feared consequence. (4) Checking (38%), which is a compulsive seeking of reassurance that a task has been done correctly by visually, physically, or verbally checking repeatedly. Commonly, patients need to be sure that a door is locked, the stove is off, the appliances are unplugged, or the windows are shut before retiring or leaving the dwelling. Checking is far beyond what is necessary to ensure that the task has been done correctly. Verbal checking entails asking others for reassurance. (5) Meticulous behavior (9%), which prompts patients to ensure that all things are in the proper place and order. (6) Avoiding (51%), a practice of attempting to prevent contact with stimuli that cause obsessive compulsive behavior. Successful avoidance may prevent patients from experiencing obsessions or compulsions but frequently prevents them from fulfilling occupational, scholastic, or social obligations. (7) Hoarding (percentage undetermined), an uncommon ritual in which patients collect unneeded things and spend considerable time sorting and organizing this material.⁸ Sometimes, huge quantities of materials are collected and facilities are procured to store them. Relatives' attempts to discard the "hoard" cause anxiety for patients.

Cognitive rituals and obsessional slowness are two additional compulsions. Cognitive rituals are mental activities performed by patients to decrease anxiety. The main difference between an obsessive thought and a cognitive ritual is that an obsession usually increases anxiety, and a cognitive ritual is done in an attempt to decrease anxiety.⁹ Obsessional slowness is an uncommon problem in which patients require extraordinary periods to complete simple tasks. Patients will frequently stop and start the task. They claim to be doing mental checking activities or cognitive rituals while performing the task.

Psychiatrists are frequently asked what the chances are that a patient with OCD will act on an obsession. Because obsessions frequently involve impulses to harm others or are gory, filthy, profane, or otherwise horrifying, it is under-

Table 2.—Common Obsessions Associated With Obsessive Compulsive Disorder

Disgusted response to bodily waste
Concern over dirt or germs
Fear of harm to self or others
Concern over toxic chemicals
Concern about becoming ill
Concern that others will become ill
"Forbidden" sexual thoughts
Fear of embarrassing acts
Somatic obsessions
Fear of losing things
Need for symmetry or exactness
Need to know or remember
Need to say or apologize
Need to count
Need to check

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standable that patients, care givers, and family members might seek predictions about the behavior of the patient. Greist and associates¹⁰ noted that people with obsessions that involve harming others rarely enact such obsessions. In their study, patients acted on their urges in only 2 of 400 instances, and 1 of these patients had a psychosis. The risk of a patient with uncomplicated OCD acting harmfully is extremely low.

Family members of persons with OCD also become involved in the compulsive behavior. A spouse may assist in checking behaviors before leaving the home "just to get it over with so we can get out of the house," or a relative may repeatedly assure a patient that a feared disaster has not occurred so that compulsive behavior can be averted. These involvements cause family or marital discord because patients with OCD are only temporarily reassured. Unless recognized, this discord can inhibit progress in behavior therapy or compliance with medications. In addition, secondary gain issues may perpetuate the obsessive compulsive sessions.

EPIDEMIOLOGIC FEATURES

OCD was once thought to be an unusual psychiatric condition with a prevalence of approximately 5 cases in 1,000 persons.¹¹ Recent studies such as the Epidemiologic Catchment Area Program have shown that the lifetime prevalence rate of OCD is 3.0%, the 6-month prevalence is 1.6%, and the annual incidence for all ages is 0.69 per 100 years of risk.¹²⁻¹⁵ Possible causes for previous underestimation of prevalence and incidence figures include the following: (1) most patients with anxiety disorders are treated by non-mental-health professionals, (2) patients with OCD may come to care givers with nonobsessional symptoms, (3) OCD commonly coexists with other psychiatric conditions,

and (4) patients with OCD are secretive about their condition; thus, it may not be addressed by mental-health professionals and researchers.¹⁵ The mean age at onset of OCD is 19.8 ± 9.6 years.^{15,16} The condition develops at a mean age of 17.5 years in male patients and a mean age of 20.8 years in female patients.¹⁷ Although a female preponderance exists for some anxiety disorders, the gender ratio for OCD is approximately 1:1.¹⁸ In a review by Marks,¹⁸ patients waited 8 to 12 years before obtaining care. Because of recent media presentations on this condition, patients may seek treatment sooner.

OCD seems to be familial. Between 21 and 25% of family members of probands with OCD also have this condition, and an additional 17% have some type of related symptoms.¹⁵ A review of studies of monozygotic twins by Rasmussen and Tsuang¹⁹ revealed a 63% concordant rate for obsessive compulsive symptoms. The fact that the rate was not 100% may indicate that environmental factors play a role in the pathogenesis.

Sociocultural factors among the epidemiologic features of OCD have been difficult to interpret. OCD has been encountered in many countries in addition to western ones; therefore, it may be a universal phenomenon. Social class of patients with OCD varies greatly.¹⁸ Some studies show a tendency for OCD to be associated with above-average social class and intelligence, but bias has been difficult to eliminate from such studies.

COMORBIDITY

Conditions associated with OCD are listed in Table 4.

Depression.—Depression frequently complicates OCD. As many as 80% of patients with OCD have dysphoric moods and 75% have current or past major depressive episodes.^{20,21} Many patients seek treatment because of depression, not obsessive compulsive symptoms. Conversely, de-

Table 3.—Common Compulsions Associated With Obsessive Compulsive Disorder

Cleaning or washing rituals
Checking on health status
Avoidance or cleaning rituals
Seeking reassurance about health
Distractions or "undoing" rituals
Avoiding public settings
Checking specific to obsessional content
Hoarding rituals
Arranging rituals
Asking repetitive questions
Making repetitive statements or apologies
Counting rituals
Checking rituals

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Table 4.—Comorbid Conditions Associated With Obsessive Compulsive Disorder

Depression
Tourette's syndrome
Other anxiety disorders
Anorexia nervosa
Obsessive compulsive personality disorder
Trichotillomania
Body dysmorphic disorder
Schizophrenia

pression is complicated by such symptoms in 23 to 28% of patients.²¹ The determination of which came first, depression or OCD, can be difficult, but attempting to distinguish whether depression or OCD is the primary illness is important. Usually, temporal relationships between the two symptom complexes must be determined. If OCD occurs first, presumably the anxiety disorder is the primary illness, and vice versa.

Gilles de la Tourette's Syndrome.—Tourette's disorder is typified by motor and vocal tics that occur several times during a day, almost every day, or intermittently throughout a 1-year period or more. The anatomic site, number, frequency, complexity, and severity of the tics change over time. The onset is usually before age 21 years.⁵ OCD and Tourette's disorder have many commonalities including increased familial risk of OCD in probands with Tourette's syndrome and similar longitudinal histories of chronicity and age at onset. Of patients with Tourette's disease, 90% have obsessive compulsive symptoms.²¹ Pitman and colleagues²² showed that 10 of 16 patients with Tourette's disorder (62.5%) also met criteria for OCD. In some patients, this syndrome evolves into conditions in which obsessive compulsive symptoms predominate.²³ Dopamine antagonists are the drug of choice for managing patients with Tourette's disorder, whereas patients with OCD are treated with serotonin uptake inhibitors (as subsequently discussed).

Other Anxiety Disorders.—OCD is classified with the anxiety disorders because of an association with anxiety and phobias. Symptoms of phobia have been found in 40 to 80.2% of patients with obsessive behavior.²¹ Mellman and Uhde²⁴ found that 27% of patients with panic disorder also had obsessive compulsive symptoms.

Anorexia Nervosa.—Many mental-health workers have noted similarities between anorexia nervosa and OCD. In anorexia nervosa, patients have an intense fear of gaining weight or of becoming fat, even when they are underweight. They also are disturbed with the weight, size, or shape of their body. Anorexia nervosa predominantly affects women. Kasvikis and co-workers²⁵ studied a series of 280 patients with OCD and found that 10.6% of the women had a history consistent with anorexia nervosa. Recent studies have

shown similar but not identical results of positron emission tomography in patients with OCD and anorexia nervosa.²⁶⁻²⁸ With both illnesses, the caudate nucleus demonstrates increased metabolic activity although differences exist in the frontal lobe findings.

Obsessive Compulsive Personality Disorder.—Whether obsessive compulsive personality disorder and OCD are related is controversial. Psychoanalytic theorists believe that obsessive compulsive personality disorder and OCD are on a continuum. In fact, similar defense mechanisms are found in these conditions. Several studies have shown that obsessive compulsive personality disorder is overrepresented among persons with OCD.^{20,29-36} Other studies, however, suggest that OCD is not related to obsessive compulsive personality disorder.^{21,32,33}

Trichotillomania.—Trichotillomania is thought to be associated with OCD. Patients with trichotillomania have an irresistible urge to pull out their hair; this behavior results in noticeable loss of hair. Tension occurs before the patient pulls out the hair, and relief of tension or gratification results from the actual hair-pulling activity. Trichotillomania is similar to OCD in that grooming behaviors are extreme. Patients with trichotillomania have responded to the antiobsessional drug clomipramine hydrochloride.³⁴

Body Dysmorphic Disorder.—Body dysmorphic disorder is also thought to be related to OCD. Patients with this disease are preoccupied with some imagined defect or malformation.⁵ The belief in the defect is not delusional but leads patients to become preoccupied (obsessed) with the perceived malformation. Case reports describe fluoxetine as effective in managing this condition.³⁵ Even though similar treatment for trichotillomania and body dysmorphic disorder leads to improvement, these conditions may not be related.

Schizophrenia.—Although some investigators have noted an association between obsessive compulsive and schizophrenic symptoms, longitudinal studies have not found an increased frequency of schizophrenia in either patients with OCD or their relatives.²¹ Fenton and McGlashan³⁶ found that 13% of 163 patients with chronic schizophrenia exhibited prominent obsessive compulsive symptoms. Thus, the overlap between OCD and schizophrenia is undirectional.

ETIOLOGIC FACTORS

Psychodynamic and behavioral theories have been developed in attempts to explain the cause of OCD; however, among psychiatrists, biologic theories predominate. A complex, interrelated series of hypotheses are being explored. All these hypotheses focus on the inhibitory neurotransmitter serotonin because serotonin uptake inhibitors (clomipramine, fluoxetine, and fluvoxamine) have proved superior to other medications as antiobsessional drugs.³⁷ Uptake of

serotonin is the most important inactivating step in neurotransmission of serotonin. By preventing uptake of serotonin, metabolism is diminished, and more serotonin is available to presynaptic and postsynaptic receptors. In one study,³⁸ a correlation was noted between the clinical effectiveness of the serotonin uptake inhibitors and low cerebrospinal fluid levels of 5-hydroxyindoleacetic acid.

The actual reason that increased concentrations of serotonin in the synaptic cleft decrease obsessional behavior is uncertain. At least three possibilities exist: (1) increased stimulation of the postsynaptic serotonin (5-HT) receptor; (2) down-regulation of the 5-HT receptor, which decreases serotonin transmission activity; and (3) increased stimulation of autoreceptors, which decreases the level of serotonin released with long-term administration of the serotonin uptake inhibitor. To study these hypotheses, Benkelfat and colleagues³⁹ administered either metergoline, a 5-HT receptor antagonist, or a placebo for 4 days to patients with OCD who were receiving clomipramine. Patients experienced a partial relapse of OCD while receiving metergoline. Thus, 5-HT receptor stimulation seems important to the antiobsessional effect of clomipramine. Paradoxically, results have been attained when the 5-HT agonist *m*-chlorophenylpiperazine (mCPP) was administered. Patients who do not receive pretreatment with clomipramine experience exacerbation of OCD. Patients with OCD who were being treated with clomipramine were challenged with mCPP after a 4-month regimen of clomipramine. Symptoms of OCD did not increase substantially.⁴⁰ One possible conclusion is that clomipramine decreases the patient's responsivity to mCPP through down-regulation of 5-HT receptors by decreasing the absolute number or sensitivity of receptors. The antiobsessional effects of serotonin uptake inhibitors are not immediate; several weeks of administration of the drug are necessary, similar to that seen with use of antidepressants for depression. In the antidepressant and depression paradigm, improvement in clinical course correlates with down-regulation of β_1 -adrenergic receptors. Although receptor down-regulation has not been proved to alleviate depression, it is strongly suggestive, and it is tempting to speculate that a similar action occurs with 5-HT receptors in patients with OCD.⁴¹

The observation that OCD occurs comorbidly with Tourette's syndrome led Goodman and co-workers²³ to hypothesize that an interaction between serotonin and dopamine neurotransmission causes either OCD or Tourette's disease. In Tourette's disorder, the dopamine system is thought to be dysfunctional because patients with this condition respond to haloperidol and other dopamine D2 antagonists.⁴² Thus, postsynaptic dopaminergic receptors may be supersensitive in Tourette's syndrome. Perhaps, serotonin, which has an inhibitory function to dopaminergic neurons,

can create a spectrum of illnesses on the basis of the degree of dysregulation of serotonin and dopamine.

Baxter⁴³ noted that von Economo's encephalitis, Meige's disease, Sydenham's chorea, and Huntington's disease have all been reported to cause OCD-like symptoms. He also noted that cingulotomy, which interrupts tracks between the caudate nucleus and orbital area to the cingulate, sometimes diminishes such symptoms. These observations along with known neuroanatomy and site of dopaminergic and serotonergic receptors led Baxter to develop a unifying hypothesis that focuses on the striatum. Accordingly, the putamen and the caudate nucleus are key elements in the neuropathologic development of OCD. These areas are rich in serotonin and dopamine receptors and receive input from the motor strip, association cortex, limbic system, and limbic prefrontal cortex.⁴⁴ Baxter proposes that pathologic features of the striatum might cause tics, OCD, and more complex behaviors that involve obsessional thoughts, depending on the site of the disorder.⁴³ This is an area of intense research. New technologies are being used, such as positron emission tomography, single photon emission computed tomography, and brain electrical activity mapping, and pharmacologic studies are being done to clarify the cause of OCD.

MANAGEMENT

Education and Self-Help.—Most patients who seek treatment of OCD will benefit from education about the condition. Patients with OCD have (1) distortions about the actual nature of this illness, (2) shame and guilt, (3) a fear of becoming insane, (4) a terrifying feeling of such uniqueness that no one understands their condition, and (5) a fear of confinement if the disorder is discovered. Many patients tell nobody of their predicament. An important step in establishing a therapeutic alliance occurs when the physician conveys a thorough understanding about OCD to the patient and educates the patient. Sometimes, patients have seen television shows or have read books about OCD before seeking help from a psychiatrist. Nonetheless, these patients are still not well educated about OCD and have distorted such information because of their own beliefs.

Family members of patients with OCD frequently engage in compulsive behavior with patients. These persons also must be educated because as part of the behavior therapy program, they need to help patients by not engaging in ritualistic behavior. Mental-health advocacy groups have been laboring to change the environment for mentally ill persons. One group, the Obsessive Compulsive Foundation, Inc. (P.O. Box 9573, New Haven, CT 06535; telephone [203] 772-0565), has done an excellent job of developing a list of mental-health workers who are interested in treating patients with OCD. The Foundation also provides a useful

newsletter for a fee. All patients with OCD should be referred to this group.

Pharmacotherapy.—Throughout the years, virtually every class of psychotropic agent has been used in an attempt to control obsessive compulsive symptoms. The only drugs that demonstrate consistent effectiveness are the serotonin uptake inhibitors.⁴⁵ This group of drugs includes clomipramine, fluoxetine, fluvoxamine, and sertraline. The last two drugs are available only for experimental protocols in the United States. Of all these agents, clomipramine has been studied the most. Numerous controlled trials have shown it to be efficacious.^{46,47} Fluoxetine may be as effective as clomipramine^{48,49} but with fewer side effects.⁴⁵ On the basis of Greist's review of the literature,⁴⁵ the usual effective dosage of clomipramine is 100 to 250 mg/day and of fluoxetine is 20 to 80 mg/day. In a study in which Pigott and associates⁴⁸ compared clomipramine with fluoxetine, the mean dosage of clomipramine was 209 ± 13 mg/day and of fluoxetine was 75 ± 4 mg/day.

The effectiveness of these agents is independent of the antidepressant activity. Six to 10 weeks may elapse before the patient responds to therapy.⁵⁰ Reduction in obsessive compulsive symptoms varies from 30 to 60%.^{45,51} In one study, discontinuation of the serotonin uptake inhibitor clomipramine led to relapse in up to 89% of patients who were treated for a mean of 10.7 ± 5.5 months.⁵²

Fluvoxamine and sertraline have also been analyzed for managing OCD. Jenike and colleagues⁵³ studied 38 patients with OCD in a 10-week, double-blind, placebo-controlled trial of fluvoxamine. A mean dosage of 294 mg/day of fluvoxamine yielded significantly better results than did placebo in two of three measures of improvement in patients with OCD. Results with sertraline have varied. Jenike and co-workers⁵⁴ reported that 10 patients in a 10-week, double-blind trial of sertraline, 200 mg/day, failed to show significant improvement in four measures of obsessive compulsive symptoms in comparison with results with placebo. Chouinard and associates,⁵⁵ who studied 78 patients with OCD without depression in a double-blind, placebo-controlled trial of the efficacy of 200 mg/day of sertraline, found improvement in three of four measures of OCD and a trend of improvement in the fourth measure.

When pharmacotherapy with any of these agents is instituted, a discussion of side effects should precede treatment. Usually, the initial dosage of clomipramine is 25 mg/day; it is gradually increased as tolerated up to 100 mg/day during the first 2 weeks. During initial titration, clomipramine is usually administered in divided doses with meals to decrease gastrointestinal side effects. Thereafter, the dosage may be increased gradually during the next several weeks, up to a maximum of 250 mg/day. Once titration of the dose is completed, the total daily dose may be administered at bed-

time to minimize daytime sedation. With fluoxetine, the initial dose is 20 mg administered in the morning (fluoxetine tends to cause insomnia if administered at bedtime). Fluoxetine has a long elimination half-life, and its full effect may be delayed as much as 4 to 6 weeks. Although it is difficult to determine how rapidly to increase this medication, most studies indicate that 60 to 80 mg/day is necessary to manage OCD. Therefore, the drug is usually increased gradually up to 60 to 80 mg/day during a 4- to 8-week period.

If these medications fail to improve a patient's condition and especially if additional behavior therapy or other management approaches (as subsequently discussed) fail, strategies of augmentation should be considered. Although lithium has been used to augment antidepressants for treating patients with depression, it has not been shown to be as effective in patients with OCD.⁴⁵ In difficult cases, however, augmentation of lithium may be indicated. Fluoxetine augmented with buspirone hydrochloride may be effective.^{51,56} Finally, antipsychotic agents may be helpful in patients with OCD who also have either schizotypal personality disorder or Tourette's syndrome. In a group of patients with OCD with these characteristics, McDougle and colleagues⁵⁷ reported that 88% failed to respond to fluvoxamine or fluvoxamine in combination with lithium but experienced improvement when haloperidol or pimozide was added. Although benzodiazepines are commonly used in conjunction with serotonin uptake inhibitors, no systematic studies of their use as augmenting agents exist. To date, these drugs seem idiosyncratically effective.⁵⁸

No studies have adequately determined the duration a patient should remain on medications. Because studies have shown a high relapse rate in patients who discontinue pharmacotherapy,⁵² patients may need medication indefinitely.

Behavior Therapy.—Behavior therapy is as important as pharmacotherapy in helping most patients manage OCD. The preferred treatment is exposure and response prevention.^{59,60} Exposure means that patients are asked to interact with stimuli that result in obsessions or ritualistic behavior. Response prevention means that patients are asked to refrain from all ritualization even though urges to ritualize may be extremely strong. For patients who have catastrophic thinking, imaginal exposure is essential. In imaginal exposure, a patient is asked to think about the catastrophic ideation intensively and to imagine that it comes true. This technique decreases the patient's need to defend against these thoughts with compulsive behavior. Without imaginal exposure, relapse occurs more frequently.⁶¹

Several variables in the behavior treatment of patients with OCD have been determined. Graded exposure to stimuli over time seems to be as effective as rapid exposure. Exposure periods of long duration are more effective than are short sessions, and in vivo exposure, when possible, is

more beneficial than imaginal exposure. Patient-directed exposures are as effective as therapist-guided exposures or modeling. Some patients, however, need assistance during the initial phase of their treatment; thereafter, they can self-direct their exposure sessions. As with all treatments, a good therapist-patient relationship is essential. Therapists, however, must not be too permissive or too tolerant because such an attitude has been associated with a poor outcome. The degree of strictness of the response prevention portion of a patient's program needed to produce improvement has not been determined; however, use of physical force is not advocated.^{59,61}

Exposure, response prevention, and, when necessary, imaginal exposure seem most effective for patients who perform physical rituals such as cleaning, repeating, completing, checking, meticulous behavior, hoarding, and avoiding. In comparison, patients who perform cognitive rituals are more difficult to treat because such rituals are not observable by therapists and they do not seem to be as controllable for the patients. Techniques such as distraction, whereby the patient engages in conversation or purposefully notices the environment, or relaxation therapy help obviate troublesome thoughts or cognitive rituals. Thought stopping and aversive therapy (self-administering punishment, such as a snap of a rubberband around the wrist) have also been described.⁶² Usually, medication offers the best hope for patients who perform cognitive rituals, inasmuch as such patients can be difficult to manage with behavior treatment. Some symptoms, including untreated depression, schizotypal personality disorder or traits, and overvalued ideation, respond poorly to behavior treatment, as do patients who take large doses of sedatives and who did not comply with previous therapy.^{45,58} Overall, in approximately 90% of patients who undergo a course of behavior therapy with exposure and response prevention, symptoms are decreased by 50% or more; however, a fourth of the patients refuse to cooperate with the proposed plan.^{45,63}

When Marks⁶⁴ reviewed the long-term follow-up of behavior management of patients with OCD, he found that improvement persisted 2 to 4½ years after completion of such therapy. Overall, the patient's condition was described as much better in 57 to 75%, as improved in 5 to 25%, and as unchanged in 20%.

When patients with OCD are successfully managed with medication or behavior therapy (or both), they have substantial free time that must be filled with some other activity. Social skills training, vocational counseling, and education in the use of leisure time are frequently necessary to help ease a patient's transition to a less symptomatic life-style.⁶²

Psychotherapy.—Psychoanalysis and dynamic psychotherapy have been unsuccessful in managing OCD.^{45,65,66} The usefulness of cognitive intervention has not been thor-

oughly assessed, and further evaluation is needed.^{67,68} Most patients with OCD seem to benefit from supportive psychotherapy. Patients who undergo behavior therapy encounter frightening stimuli. The response prevention part of the program can also be terrifying to patients, especially if they have catastrophic thinking. Support of the patient is part of behavior therapy and does not interfere with the patient's ability to improve. Moreover, support may decrease the dropout rate and increase compliance. Therapists must be cautious not to engage in verbal checking with patients. This situation occurs when a patient requests repeated reassurances that it is safe to avoid performing a ritual or to ignore an obsession.

Psychosurgical Intervention.—Psychosurgical intervention should be considered only for patients who do not respond to any other therapy or who have disabling symptoms.^{69,70} Since the origin of psychosurgical treatment, various techniques have been used, including creation of lesions in the orbital frontal cortex, anterior cingulum, and posterior orbital cortex. With use of modern-day stereotactic cingulotomy procedures, adverse effects are rare, and personality is usually unaffected; approximately 75% of patients experience improvement.⁷⁰

Integration of Treatment.—Greist⁵⁸ has developed a decision tree for the treatment of patients with OCD (Fig. 1). When the disorder is diagnosed and the patient is dysfunctional, therapy should be initiated. The patient's condition and available resources as well as preference of medications or other treatment options are helpful in determining appropriate initial therapy. Refinement of the behavior therapy and changes in pharmacotherapy with regard to use of augmentation agents or different serotonin uptake inhibitors depend on the patient's response. Finally, if all other options fail, psychosurgical intervention can be considered for severe disorders.

CONCLUSION

OCD has been described as a severe and disabling anxiety disorder. Fortunately, with the advent of serotonin uptake inhibitors and refined behavior therapy techniques, patients with OCD have an excellent chance for reduction of symptoms. Future research should lead to increased understanding of the cause of the condition and improved treatment.

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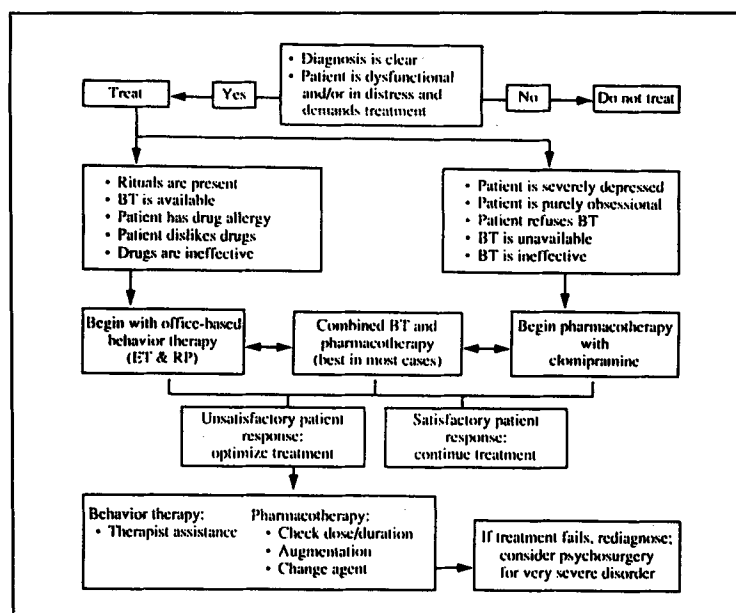


Fig. 1. Diagram of decision tree for treatment of patients with obsessive compulsive disorder. BT = behavior therapy; ET = exposure therapy; RP = response prevention. (From Greist.⁵⁸ By permission of Physicians Postgraduate Press, Inc.)

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